

Spring

The Boundaries of Spring:
Mathematical Limits of Auditability and Non-Auditability

SCX

202671

v1.0 — — SCX — Spring

Abstract

SPRING—— SPRING (1) **Hoeffding**——Spring (2) **Gödel**——Spring (3) ≠Korzybski——Spring
Boundary Theorem SpringSpring $1 - e^{-2M\Delta^2}$ ——Spring Spring
Spring—— Spring GPTSpring
SpringHoeffdingGödel- SCX

Contents

1

1.1

"""capability
 SPRING——SPRING

□

1.2 Spring

SPRING [?]" Training-Is-Audit GPTClaudeGeminiM = 1UNDECLARED SPRING——M_t
 CERCISPRING**born audited**——
 ——Hoeffding——
 SPRING

1.3

- 1 **Hoeffding** $\mathbb{P}(\text{error}) \leq e^{-2M\Delta^2}$ M → ∞00 Spring——Spring
 - 2 **Gödel** Spring Gödel Spring——M
 - 3 ≠ DFTYAJIE = max13 Spring
- Hoeffding, 1963 [?] Gödel, 1931 [?] Korzybski, 1933 [?] Spring—— SpringSpring

1.4

Boundary Theorem

Definition 1.1 (Spring). $\mathcal{F}_{\text{SPRING}} \textit{Spring}——\textit{Spring}$

- Auditable(SPRING) = { $\varphi \in \mathcal{F}_{\text{SPRING}}$ | SPRING φ }
- (SPRING) = $\mathcal{F}_{\text{SPRING}}^c \cup \{\varphi \in \mathcal{F}_{\text{SPRING}} \mid \varphi\}$

∂_{Spring} SPRING——

——**HoeffdingGödelKorzybski**

1.5

- (1) 21——Hoeffding
- (2) 32——Gödel
- (3) 43—— $\neq$
- (4) 5——
- (5) 6Spring——
- (6) 7Spring——
- (7) 8——GPT/Claude/Gemini
- (8) 9——

Corollary 2.3 (Spring). $\text{SPRING confidence interval absolute guarantee} \text{---Hoeffding}$

$$\boxed{\text{Spring} X \pm \sqrt{1 - e^{-2M\Delta^2}} X}$$

GPT-500K --- $\text{SPRING} + M + \Delta + \text{''''''}$ ---
SpringSpringGPT''' Spring --- Hoeffding

2.4

100%

- (1) $M = 20\Delta = 0.3 \quad e^{-2 \cdot 20 \cdot 0.09} = e^{-3.6} \approx 0.027 \text{---} 97.3\% \text{ ''}$
- (2) 97.3%SPRING97.3% Hoeffding $M\Delta$
- (3) ---SPRING100% bug SPRING

3 2Gödel

3.1

1—— SPRING SPRINGYAJIE ””””
 SPRING SPRINGGödel

3.2

SPRING

Definition 3.1 (Spring). $\mathcal{F}_{\text{SPRING}}$ SPRING

- DFT
- $\wedge, \vee, \neg, \implies$
- $\forall \exists$
- $E_1 < E_2 \| \mathbf{f}_1 \| < \| \mathbf{f}_2 \|$

SPRING

$$\mathcal{A}_{\text{SPRING}} : \mathcal{F}_{\text{SPRING}} \rightarrow \{0, 1, UNDECIDED\}, \quad (6)$$

$$0 \text{ ” ” } 1 \text{ ” ” } \geq 1 - e^{-2M\Delta^2} UNDECIDED \text{ ” ” }$$

Remark 3.2 ($\mathcal{F}_{\text{SPRING}}$). $\mathcal{F}_{\text{SPRING}} \text{ F}_{\text{SPRING}} || \forall E < E'$
 $\mathcal{F}_{\text{SPRING}} \text{ Gödel ” ” }$

3.3

Theorem 3.3 (SpringGödel). $\mathcal{F}_{\text{SPRING}} \varphi_{\text{SPRING}} \in \mathcal{F}_{\text{SPRING}}$

(i) $\varphi_{\text{SPRING}} \varphi_{\text{SPRING}}$

(ii) $\mathcal{A}_{\text{SPRING}}(\varphi_{\text{SPRING}}) = UNDECIDED \text{—— SPRING } \varphi_{\text{SPRING}} M$

$\varphi_{\text{SPRING}} \text{—— SPRING } Gödel$

. $[Gödel \text{ [?]} \mathcal{F}_{\text{SPRING}}$

1 $\mathcal{F}_{\text{SPRING}} \text{ Gödel DFTZr } \mathbf{a}_i \text{ YAJIE}$

2 $\mathcal{F}_{\text{SPRING}}$

$$\text{Provable}_{\text{SPRING}}(x) \iff x \mathcal{F}_{\text{SPRING}} \text{ YAJIE} \quad (7)$$

diagonal lemma [?]

3
Gödel

$$\varphi_{\text{SPRING}} \equiv \text{“} \varphi_{\text{SPRING}} \mathcal{F}_{\text{SPRING}} \text{”} \tag{8}$$

$$\varphi_{\text{SPRING}} \equiv \neg \text{Provable}_{\text{SPRING}}(\lceil \varphi_{\text{SPRING}} \rceil)$$

4

- $\mathcal{A}_{\text{SPRING}}(\varphi_{\text{SPRING}}) = 1 \text{ Provable}_{\text{SPRING}}(\lceil \varphi_{\text{SPRING}} \rceil) \neg \varphi_{\text{SPRING}} \text{——} \varphi_{\text{SPRING}}$
 - $\mathcal{A}_{\text{SPRING}}(\varphi_{\text{SPRING}}) \neq 1 \neg \text{Provable}_{\text{SPRING}}(\lceil \varphi_{\text{SPRING}} \rceil) \text{——} \varphi_{\text{SPRING}} \mathcal{A}_{\text{SPRING}}(\varphi_{\text{SPRING}}) 10 \text{——}$
- UNDECIDED

5
Gödel

$$\text{“} V_{\theta} \mathbf{x} \mid V_{\theta}(\mathbf{x}) - E_{\text{DFT}}(\mathbf{x}) \mid < \varepsilon \text{ } V_{\theta} \text{SPRING} M = 100 \text{” } \text{SPRING } \text{Gödel} \mathcal{F}_{\text{SPRING}} \tag{\square}$$

3.4

Corollary 3.4. ———SPRING———

SPRING *Gödel*

SPRING

- **DFT** Kohn-Sham-
-
-
- $\varepsilon \varepsilon 3$

$\mathbf{g}_{\text{SPRING}} = \mathbf{0}$ SPRING””SCX3 UNDECIDED

GödelSpring—— SpringGPTGPT

4 3≠

4.1 Korzybski

Alfred Korzybski1933*Science and Sanity* [?]

The map is not the territory.

 □

SPRINGSPRING

4.2 Spring

SPRING

1 DFT-PBE [?]*SCAN* [?]- ———

2

3 $\text{SPRING}V_{\theta}(\mathbf{x})$ Born-Oppenheimer ———SPRING

4 k

$\varepsilon_{\text{phys}}$ **Spring**

4.3 M'''

$y^*\hat{y} = y^* + \varepsilon_{\text{phys}}$

$MV_{\theta}^{(i)}$ DFT prediction $\hat{y}y^*$

$$\mathbb{E}[V_{\theta}^{(i)}(\mathbf{x})] = \hat{y}(\mathbf{x}) = y^*(\mathbf{x}) + \varepsilon_{\text{phys}}(\mathbf{x}), \tag{9}$$

$\varepsilon_{\text{phys}}(\mathbf{x})$ ———— M

4.4

3

Theorem 4.1 ($\neq$). $y^*\hat{y}V_{\text{consensus}}$ YAJIE SPRING

$$\varepsilon = \max(\underbrace{\varepsilon_{stat}(M, \Delta)}_1, \underbrace{\varepsilon_{phys}}_3), \tag{10}$$

- $\varepsilon_{stat}(M, \Delta)$ *Hoeffding* $1M \rightarrow \infty \varepsilon_{stat} \rightarrow 0$
- ε_{phys} $\mathfrak{I}\varepsilon_{phys}M$ *DFT*

$$\varepsilon_{phys} > 0$$

$$\lim_{M \rightarrow \infty} \varepsilon = \varepsilon_{phys} > 0. \tag{11}$$

$$\begin{aligned} & \cdot \quad \boxed{1} \quad 1 \varepsilon_{\text{stat}} \leq e^{-2M\Delta^2} \rightarrow 0 \\ & \quad \quad \quad \boxed{2} \quad \varepsilon_{\text{phys}} = \sup_{\mathbf{x}} |\hat{y}(\mathbf{x}) - y^*(\mathbf{x})| \quad \text{DFT} \\ & \quad \quad \quad \text{-----} \\ & \quad \quad \quad \boxed{3} \quad \text{YAJIE } M \text{YAJIE} \text{-----} \quad \varepsilon_{\text{phys}} \\ & \quad \quad \quad \boxed{4} \quad \varepsilon = \max(\varepsilon_{\text{stat}}, \varepsilon_{\text{phys}}) \quad \varepsilon_{\text{stat}} \rightarrow 0 \varepsilon_{\text{phys}} \quad \square \end{aligned}$$

4.5 Corollary: Spring

Corollary 4.2. SPRING

- SPRINGDFT PBE””
- SPRING ””
- SPRING ””””

——SPRING

SpringDFT—— ”AlNa=3.11Å”——Spring12=0.004Å 98.7%SpringDFT PBEAlN
Springground truthDFT PBE ””””””””

5

5.1

123

Theorem 5.1 (Spring —). $\mathcal{F}_{\text{Spring}} \text{Spring} \mathcal{A}_{\text{Spring}} \text{Spring SpringAuditable}(\text{Spring})(\text{Spring})$
 ∂_{Spring}

$$1 \text{ Hoeffding } \varphi \in \mathcal{F}_{\text{Spring}} \text{Spring} 1 - e^{-2M\Delta^2} M\Delta - 1MM$$

$$\forall M < \infty : \mathbb{P}(\mathcal{A}_{\text{Spring}}(\varphi)) > 0. \quad (12)$$

$$2 \text{ Gödel } \varphi_{\text{Spring}}^{\perp} \in \mathcal{F}_{\text{Spring}} \mathcal{A}_{\text{Spring}}(\varphi_{\text{Spring}}^{\perp}) = \text{UNDECIDED Spring} \text{--- Spring}$$

$$\exists \varphi \in \mathcal{F}_{\text{Spring}} : (\varphi) \wedge (\mathcal{A}_{\text{Spring}}(\varphi) = \text{UNDECIDED}). \quad (13)$$

$$3 \text{ Korzybski } \text{Spring} \varepsilon \geq \varepsilon_{\text{phys}} \varepsilon_{\text{phys}} \text{Spring}$$

$$\lim_{M \rightarrow \infty} \varepsilon = \varepsilon_{\text{phys}} \not\rightarrow 0 \varepsilon_{\text{phys}} = 0. \quad (14)$$

$\partial_{\text{Spring}} \text{Spring}$

- $\text{Spring} 1 - e^{-2M\Delta^2}$
- $\text{Gödel Spring UNDECIDED} \text{---}$

5.2

Self-Awareness of the Boundary

Definition 5.2. $\text{Spring} \varphi \text{Spring}$

$$(1) \varphi \mathcal{F}_{\text{Spring}}$$

$$(2) \varphi \Delta M$$

$$(3) \varphi$$

$$(4)$$

Proposition 5.3. Spring

- $\mathcal{F}_{\text{Spring}} \text{---}$
- $\text{Hoeffding} M\Delta \text{---} \Delta M \text{Spring}$
- $\text{Spring "DFT"" --- Spring}$

Spring""GPT"" — "" /"" Spring"76%" — 76%100%

5.3

SPRING

Protocol 5.4. SPRING $\text{---}\Delta M\text{---}$

- (1) $\theta_{\min} 95\%$ UNDECIDED
- (2) "Spring [M//]"
- (3) DFTCCSD(T)
- (4) ---Spring

6 Spring

6.1

SPRING SPRING——

SPRING

(1) $\mathcal{F}_{\text{Spring}}$ SPRING—— DFT

(2) SPRING/ $\Delta > 0$

(3) SPRING ——DFT PBEDFT PBE

6.2

SPRING

1 SPRING M $M\delta$ HoeffdingRMSE

$\mathbf{x}$

$$\mathbb{P}(|V_{\text{consensus}}(\mathbf{x}) - \mathbb{E}[V(\mathbf{x})]| > \delta) \leq e^{-2M\Delta^2}. \quad (15)$$

2 SPRINGYAJIE SCX1M

$$\eta F1 \geq 1 - \frac{1}{\eta} \sum_s \rho_s e^{-2M\Delta_s^2}$$

3 SPRING ”(111)”——DFT(111)

4 SPRING””Query-Is-Computation ””—— SPRINGLLMSpring M_t

5 SPRINGCERCIS CERCISYAJIE M_t

””Hoeffding

6 - SPRING $\mathcal{H}(\mathcal{D}) = \mathcal{H}(\mathcal{D})$

$\mathcal{H}(\mathcal{D})$

6.3

Table 1: Spring

/	Hoeffding	1
	F1	1+3
		2
		—
		1
-		—

”” ”” _____

7 Spring

7.1

SPRINGSPRING—— SPRING

7.2

1 SPRINGDFT"" SPRINGDFT

3——≠ SPRINGDFT ""——SPRING
UNDECIDED +

2 SPRINGACE [?]MACE [?] NequIP [?]""

2Gödel+ 3 ——
UNDECIDED +

3 Gödel SPRING—— ”SpringM”

2——Gödel SpringSpring
UNDECIDED + Gödel

4 SPRING100%

1——Hoeffding $M0$ ——Spring
 $1 - e^{-2M\Delta^2} < 100\% + 100\%$

5 SPRING $\mathcal{F}_{\text{SPRING}}$ ——

Spring Spring
OUT OF SCOPE——”Spring”

6 SPRING—— $\mathcal{F}_{\text{SPRING}}$

Δ - Hoeffding
INSUFFICIENT DATA + ΔM

7 SPRING——

13 ——
SPECULATIVE +

8 SPRING——

2 + 3——DFT Spring""""
+ ”[]”

7.3

Table 2: Spring

Gödel	3	UNDECIDED	/
	2+3	UNDECIDED	/
	2	UNDECIDED	
	1	¡100%	——
		OUT OF SCOPE	
	1	INSUFFICIENT DATA	
	1+3	SPECULATIVE	
	2+3		

Table 3: *
.....

::

8 GPTClaudeGemini

8.1 vs.

SPRINGGPT [?]Claude [?]Gemini [?] ———
””

- (1) $M = 1$ SCX2 GPT””——GPTGPT
- (2) UNDECLARED —— $M\Delta$
- (3) GPTClaudeGemini—— ”PythonCSV” ”” ””””
- (4)
_____””

8.2

??

Table 4: Spring vs. —

	Spring	GPT/Claude/Gemini	
	$M \geq 8$	$M = 1$	vs.
	AUDITED	UNDECLARED	vs.
	Hoeffding	””	vs.
			vs.
	$\mathcal{F}_{\text{SPRING}}$		vs.
Gödel			vs.
	UNDECIDED		vs.
/			vs.
	+		vs.
			vs.

8.3

”AIN500K”

GPT/Claude/Gemini

"AlN500Ka = 3.114 Å c = 4.986 Å "

-
- """"
-
-

Spring

AlN500Ka = 3.115 ± 0.003 Å, c = 4.991 ± 0.005 Å

- YAJIE $M = 12$
 - 97.3% (Hoeffding $e^{-2 \cdot 12 \cdot 0.15^2} \approx 0.027$)
 - CERCIS0.91/1.0
 - DFT PBE + SpringACE
 - $-\mathcal{H}(\mathcal{D}) = \text{a3f2...e71b}$
 - 12- $\rightarrow \rightarrow$ 500K
 - DFT PBE $\varepsilon_{\text{phys}}$ PBE vs.
 - +
 - 12
 - 97.3%Hoeffding
 - DFT PBE
 - $\varepsilon_{\text{phys}}$
- GPT Spring

8.4

- (1) GPTtoken——
- (2) ——
- (3) GPT"""" ——GPT """" —— ""
- (4) "" GPT"""" ——
""hallucination

——

9

9.1

SPRING

1 Hoeffding SPRING

2 Gödel SPRINGSPRING SPRING

3 $\neq$ SPRING

SPRING

SPRING $1 - e^{-2M\Delta^2}$ SPRING

9.2

□

——Hoeffding

SPRING

SPRINGBoundary Self-Awareness—— ————

9.3 GödelSpring

Kurt Gödel1931

Hilbert

Gödel—— ”” ””

SPRINGGödel——HoeffdingKorzybski——

SPRING—— SPRING

Gödel:

Spring: ——

9.4

SPRINGAI

(1)

””””——GPT””””

(2)

—— SPRING

(3) ———SPRINGAI——— ”” ””limitations section ———7

(4) ——— ——— ”” ””

9.5 SpringSpring

Alan Turing—— - ——

SPRINGSPRING—— SPRING SPRING

SpringSpring

□

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Hoeffding (1963) [?] Gödel (1931) [?]Korzybski (1933) [?] ””——””

SPRING

SCX””Theorem 3””Theorem 2——

SCX

SCX

SCX[repository] SPRING5

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A A

A.1 Spring

Definition A.1 (Spring —). $\text{SPRING}\mathcal{A}_{\text{SPRING}} = (\mathcal{F}_{\text{SPRING}}, \mathcal{E}, \mathcal{P}, \mathcal{C}, \mathcal{B})$

- $\mathcal{F}_{\text{SPRING}}$
- $\mathcal{E} = \{E_1, \dots, E_M\}M$
- $\mathcal{P}\text{YAJIE}$
- $\mathcal{C}\text{CERCISM}_t$
- $\mathcal{B}$

A.2 Hoeffding

Definition A.2 (Hoeffding). $\varphi \in \mathcal{F}_{\text{SPRING}} \ C_i(\varphi) \in [0, 1]E_i\varphi$

$$\bar{C}(\varphi) = \frac{1}{M} \sum_{i=1}^M C_i(\varphi). \quad (16)$$

$\varepsilon > 0$

$$\mathbb{P}(|\bar{C}(\varphi) - \mathbb{E}[\bar{C}(\varphi)]| \geq \varepsilon) \leq 2 \exp(-2M\varepsilon^2). \quad (17)$$

A.3 Gödel

$\mathcal{F}_{\text{SPRING}}\text{Gödel} \text{---}$

Example A.3 (SpringGödel). (1) $\text{AuditPass}_M(x, \varphi) \text{ ”} xM \geq 1 - e^{-2M\Delta^2}\varphi \text{”}$

$$(2) \ d(\varphi) = \varphi(\lceil \varphi \rceil)$$

$$(3) \ \varphi_{\text{SPRING}} \equiv \neg \text{AuditPass}_M(\lceil \varphi_{\text{SPRING}} \rceil, \varphi_{\text{SPRING}})$$

$$\varphi_{\text{SPRING}} \text{ ”Spring” } \varphi_{\text{SPRING}} \text{AuditPass}_M(\lceil \varphi_{\text{SPRING}} \rceil, \varphi_{\text{SPRING}}) \ \varphi_{\text{SPRING}} \ \mathcal{A}_{\text{SPRING}}(\varphi_{\text{SPRING}}) = \text{UNDECIDED}$$

A.4 -

Definition A.4 ($\varepsilon_{\text{phys}}$). y

- $y^*y \text{---}$
- $\hat{y}_{DFT}y_{DFT}$
- $\varepsilon_{\text{phys}} = \sup |\hat{y}_{DFT} - y^*|$

$$\text{Spring}\hat{y}_{DFT} \ \mathbb{E}[V_{\theta}^{(i)}] = \hat{y}_{DFT} \neq y^* \varepsilon_{\text{phys}} > 0$$

B B

	English
SPRING	Spring is the most auditable framework ever built.
SPRING	Precisely because it is rigorous, Spring can identify its own fundamental limits.
— M Spring	The audit error is strictly positive — for any finite M , knowledge under Spring is always probabilistic.
Spring	Spring can give confidence intervals, not guarantees.
—	This is not a flaw — it is the mathematical nature of empirical knowledge.
	Any sufficiently strong formal system contains true but unprovable propositions.
Spring—	Spring must declare its axiomatic boundaries — which axioms it assumes, which it cannot prove.
	The map is not the territory.
$M\varepsilon$ YAJIE— ε	If all M experts agree within approximation error ε , Yajie returns HIGH confidence — but the territory may differ from the map by up to ε .
$= \max(1, 3)$	The audit error $= \max(\text{statistical error from Ceiling 1, systematic error from physical approximation})$.
SpringSpring $1 - e^{-2M\Delta^2}$	For claims within Spring's formal system, Spring provides an audit with confidence $1 - e^{-2M\Delta^2}$.

	English
Spring	For claims outside the formal system, Spring is silent.
Spring	The boundary is self-aware: Spring knows what it can audit, what it cannot, and why.
GPTSpring	GPT doesn't know its own limits. Spring does. That is the difference between a model and an audit framework.
	The strongest framework is not the one that claims it can audit everything — it is the one that knows precisely what it cannot audit.
Gödel:	Gödel: This is true, but I cannot prove it.
Spring: —	Spring: This may be true, but I cannot audit it — here is why.
	Silence is a capability, not a flaw.
	A system operating without self-awareness is blind. A system operating with boundary self-awareness is awake.
Spring	Spring audits what it can audit, and precisely declares what it cannot. That is the entire difference between an audit framework and a hallucination generator.

Table 5: Spring